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Preparing Magnon–Photon Pairs in a Cavity–Magnon–Qubit System

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Hybrid quantum states consisting of single photons and single magnons are valuable quantum resources for realizing multiple quantum information tasks. The blockade effects of magnons and photons in a cavity–magnon–qubit system are investigated, where an auxiliary microwave cavity is simultaneously coupled with an yttrium iron garnet sphere, a qubit, and another microwave cavity. The modes of the auxiliary cavity are adiabatically eliminated, and the virtual photons in the auxiliary cavity mediate the coupling between the magnon mode and photon mode. When the system parameters satisfy both conventional and unconventional blockade conditions, both the magnon/photon auto-correlation and magnon–photon cross-correlation are weak, indicating that magnons and photons tend to exist alone in the system. When the system parameters only satisfy unconventional blockade conditions, strong magnon–photon cross-correlation can be exhibited, indicating that magnons and photons tend to exist simultaneously in the system. The proposal presents a method for preparing magnon–photon pairs, which may have potential applications in quantum information sciences and technologies.

1. Introduction

In various magnon-based hybrid systems,[1,2] the coupling of magnons with optical photons,[3,4] microwave photons,[5–10] phonons,[11] and superconducting qubits[12–14] provides a wealth of options for processing quantum information and realizing integrated quantum devices.[15,16] Magnons are collective spin excitations in magnetic order materials such as yttrium iron garnet (YIG).[17,18] In YIG materials with high spin density and low damping rate, microwave photons can strongly couple with magnons. Such magnon-based hybrid quantum systems enable the realization of quantum effects of magnons, such as the ground-state cooling of a magnomechanical resonator,[19,20] magnon blockade,[21] bipartite or tripartite entanglement,[22–24] squeezed state,[25,26] and nonreciprocity.[27] Moreover, the unique advantages of magnons as quantum information carriers include their long lifetimes and high tunability of frequency.[28]

Photon blockade effect is an effective way to prepare photon number states. When photon blockade occurs in an optical cavity, the first photon excited in the cavity will suppress the excitation of the next photon, so that at most one photon exists in the system. The earliest proposed photon blockade mechanism[29–33] relies on the anharmonicity of the system energy level. When a single-photon state in the system is resonantly excited, the state with a higher number of photons is suppressed by detuning, and the system prepares as many single-photon states as possible. This usually works well in optical cavities with strong Kerr nonlinearity. Later, another type of photon blockade mechanism[34–43] that uses transitions through different paths to achieve quantum destructive interference and thus suppress high photon number states was proposed. These two mechanisms are usually referred to as conventional photon blockade and unconventional photon blockade. Furthermore, with the addition of nonreciprocal effects, many nonreciprocal photon blockade[44–46] schemes have been proposed. Magnon blockade[21,47–60] and phonon blockade[61–63] are similar to photon blockade and are the applications of the photon blockade effect in magnon and phonon modes, respectively. In addition to the preparation of single magnon states, theoretical schemes such as the preparation of magnon pair,[64,65] nonreciprocal magnon blockade,[66–69] and cavity magnon polaritons[70] have also been proposed.

Hybrid quantum states can be employed for accomplishing various quantum information tasks. For instance, correlated photon–phonon pairs can be used to implement photon–phonon quantum networks.[71] Similarly, magnon–photon pairs can offer potential value for implementing hybrid quantum networks composed of magnons and photons. Recently, the simultaneous photon–phonon–magnon blockade has been proposed in a system where a two-level atom and a YIG sphere are coupled to an optomechanical cavity.[72] By utilizing the Mollow dynamics process, entangled photon–magnon bundle emission[73] and hybrid magnon–photon bundle emission[74] have also been proposed in different hybrid cavity magnetic systems, respectively. When a magnon and a photon are emitted simultaneously, it means that there is a strong cross-correlation between the magnon and the photon. However, when the magnon blockade and the photon blockade occur simultaneously, the cross-correlation

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between magnons and photons may not be particularly strong. Therefore, in order to prepare magnon–photon pairs through the blockade effect, in addition to analyzing the conditions under which the blockade occurs, it is also necessary to analyze the cross-correlation. This method is similar to the preparation of photon pairs[31,33] or magnon pairs.

Here, we discuss a hybrid quantum system in which an auxiliary microwave cavity B is simultaneously coupled with a YIG sphere, a qubit, and a microwave cavity A. By using the adiabatic approximation to eliminate the cavity B mode, the virtual photons in cavity B mediate the indirect coupling between the cavity A mode, the magnon mode, and the qubit. We note that when there is strong coherent coupling between a qubit and a YIG sphere, applying external drive to both of them will obtain a very strong photon blockade effect,[52] which is due to the important role played by both conventional magnon blockade and unconventional magnon blockade. Therefore, we apply external drive to all three parts of the scheme we considered. The results show that qubit drive has a great influence on the blockade conditions. Under certain parameter conditions, the change of qubit drive can make the conventional blockade and unconventional blockade conditions coincide, thus showing a strong blockade effect. Under other parameter conditions, the addition of qubit drive can make both magnons and photons achieve unconventional blockade and show strong magnon–photon cross-correlation. Although this increases the complexity of the system, its effect of improving magnon–photon pairs is obvious.

2. Model and Hamiltonian

As shown in **Figure 1**a, a YIG sphere and a superconducting qubit are placed in the auxiliary microwave cavity B, and the coupling strengths with the cavity field are g_{bm} and g_{bq}, respectively. The auxiliary cavity B is coupled to the cavity A with a coupling strength of g_{ab}. The total Hamiltonian of the system is written as ($\hbar = 1$)

$$\hat{H} = \hat{H}_0 + \hat{H}_I + \hat{H}_{\mathrm{d}} \tag{1}$$

where

$$\hat{H}_0 = \omega_{\mathrm{a}}\hat{a}^\dagger\hat{a} + \omega_{\mathrm{b}}\hat{b}^\dagger\hat{b} + \omega_{\mathrm{m}}\hat{m}^\dagger\hat{m} + \omega_{\mathrm{q}}\hat{\sigma}_+\hat{\sigma}_- \tag{2}$$

$$\hat{H}_I = g_{\mathrm{ab}}\hat{a}^\dagger\hat{b} + g_{\mathrm{bm}}\hat{b}^\dagger\hat{m} + g_{\mathrm{bq}}\hat{b}^\dagger\hat{\sigma}_- + \mathrm{H.c.} \tag{3}$$

$$\hat{H}_{\mathrm{d}} = E_{\mathrm{a}}e^{-i\omega_d t}\hat{a}^\dagger + E_{\mathrm{m}}e^{-i\omega_d t}\hat{m}^\dagger + \Omega e^{-i\omega_d t}\hat{\sigma}_+ + \mathrm{H.c.} \tag{4}$$

Here, $\hat{H}_0$, $\hat{H}_{\mathrm{I}}$, and $\hat{H}_{\mathrm{d}}$ are the free term, interaction term, and driving term of the system Hamiltonian, respectively. $\hat{a}^\dagger$ ($\hat{a}$), $\hat{b}^\dagger$ ($\hat{b}$), and $\hat{m}^\dagger$ ($\hat{m}$) denote the creation (annihilation) operators of the cavity A, cavity B, and Kittel mode[75] with resonance frequencies ω_{a}, ω_{b}, and ω_{m}, respectively. $\hat{\sigma}_+$ ($\hat{\sigma}_-$) is the raising (lowering) operator of the transmon qubit with frequency ω_{q}. We consider applying microwaves[76] to drive the cavity A, the YIG sphere, and the qubit, with driving strengths E_{a}, E_{m}, and Ω, respectively, and the driving frequency is ω_{d}.

The time dependence of the Hamiltonian can be removed in a rotating frame at the driving frequency ω_{d}. The free term and the driving term in the Hamiltonian become

$$\hat{H}_0' = \Delta_{\mathrm{a}}'\hat{a}^\dagger\hat{a} + \Delta_{\mathrm{b}}'\hat{b}^\dagger\hat{b} + \Delta_{\mathrm{m}}'\hat{m}^\dagger\hat{m} + \Delta_{\mathrm{q}}'\hat{\sigma}_+\hat{\sigma}_- \tag{5}$$

$$\hat{H}_{\mathrm{d}}' = E_{\mathrm{a}}\hat{a}^\dagger + E_{\mathrm{m}}\hat{m}^\dagger + \Omega\hat{\sigma}_+ + \mathrm{H.c.} \tag{6}$$

where $\Delta_{\mathrm{a}}' = \omega_{\mathrm{a}} - \omega_{\mathrm{d}}$, $\Delta_{\mathrm{b}}' = \omega_{\mathrm{b}} - \omega_{\mathrm{d}}$, $\Delta_0' = \omega_{\mathrm{m}} - \omega_{\mathrm{d}}$, and $\Delta_{\mathrm{q}}' = \omega_{\mathrm{q}} - \omega_{\mathrm{d}}$ define the frequency detunings between the driving field and the cavity A, the cavity B, the magnon, and the qubit, respectively. The interaction part is $\hat{H}_{\mathrm{I}}' = \hat{H}_{\mathrm{I}}$.

In the large detuning limit, i.e., $\Delta_{\mathrm{b}}' \gg \{\Delta_{\mathrm{a}}', \Delta_{\mathrm{m}}', \Delta_{\mathrm{q}}', g_{\mathrm{ab}}, g_{\mathrm{bm}}, g_{\mathrm{bq}}, E_{\mathrm{a}}, E_{\mathrm{m}}, \Omega\}$,[74,77] the cavity B mode is adiabatically eliminated, thereby enabling indirect coherent coupling among cavity A, the YIG sphere, and the qubit. The effective coupling strengths of photon–magnon, photon–qubit, and magnon–qubit are given by $g_{\mathrm{am}} = -g_{\mathrm{ab}}g_{\mathrm{bm}}/\Delta_{\mathrm{b}}'$, $g_{\mathrm{aq}} = -g_{\mathrm{ab}}g_{\mathrm{bq}}/\Delta_b'$, and $g_{\mathrm{mq}} = -g_{\mathrm{bm}}g_{\mathrm{bq}}/\Delta_{\mathrm{b}}'$, respectively. Furthermore, the effective detunings of cavity

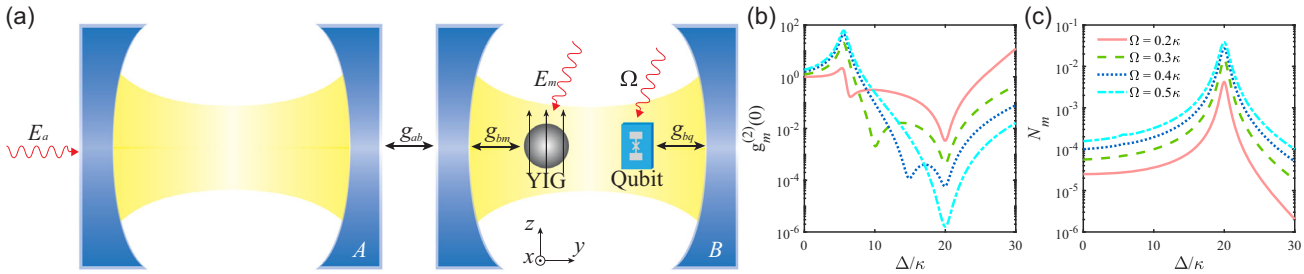

Figure 1. Sketch of the cavity–magnon–qubit system scheme in figure a). The auxiliary microwave cavity B is coupled to the YIG sphere, superconducting qubit, and microwave cavity A simultaneously, and the coupling strengths are g_{bm}, g_{bq}, and g_{ab}, respectively. The YIG sphere is in a static magnetic field with a bias direction along the positive direction of the z-axis, and the frequency of the Kittel mode in the YIG sphere is controlled by the static magnetic field. The qubit can be regarded as a two-level system containing a ground state $|g\rangle$ and an excited state $|e\rangle$. By using the adiabatic approximation to eliminate the cavity B mode, the virtual photons in cavity B mediate the indirect coupling between the cavity A mode, the magnon mode, and the qubit. In addition, the cavity A, the YIG sphere, and the qubit are all driven by external fields, with driving strengths of E_{a}, E_{m}, and Ω, respectively. The second-order correlation function $g_{\mathrm{m}}^{(2)}(0)$ of the magnon and the average number of magnons N_{m} as a function of detuning Δ are shown in b) and c), respectively. The dissipation, coupling strength, and driving strength are fixed at $\kappa/2\pi = 1$ MHz, $g_0 = 20\kappa$, and $E = 0.1\kappa$, respectively. The driving strength of the qubit considers four cases $\Omega = \{0.2\kappa, 0.3\kappa, 0.4\kappa, 0.5\kappa\}$.

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A, the magnon, and the qubit are $\Delta_a = \Delta'_a - g^2_{ab}/\Delta'_b$, and $\Delta_q = \Delta'_q - g^2_{bq}/\Delta'_b$, respectively. The effective Hamiltonian is

$$\begin{aligned}\hat{H}_{\text{eff}} &= \Delta_a \hat{a}^\dagger \hat{a} + \Delta_m \hat{m}^\dagger \hat{m} + \Delta_q \hat{\sigma}_+ \hat{\sigma}_- \\ &+ (g_{am} \hat{a}^\dagger \hat{m} + g_{aq} \hat{a}^\dagger \hat{\sigma}_- + g_{mq} \hat{m}^\dagger \hat{\sigma}_- \\ &+ E_a \hat{a}^\dagger + E_m \hat{m}^\dagger + \Omega \hat{\sigma}_+ + \text{H.c.})\end{aligned} \tag{7}$$

These dissipation processes of the system can be described by the effective non-Hermitian Hamiltonian: $\hat{H}' = \hat{H}_{\text{eff}} - i\frac{\kappa_a}{2}\hat{a}^\dagger \hat{a} - i\frac{\kappa_m}{2}\hat{m}^\dagger \hat{m} - i\frac{\gamma}{2}\hat{\sigma}_+ \hat{\sigma}_-$, where κ_a, κ_m, and γ represent the dissipation rates for the cavity A, the magnon, and the qubit, respectively. In steady state, the purity of a single photon or magnon generated in the system can be characterized by the equal-time second-order auto-correlation function $g^{(2)}_a(0) = \langle \hat{a}^{\dagger 2} \hat{a}^2 \rangle / \langle \hat{a}^\dagger \hat{a} \rangle^2$ and $g^{(2)}_m(0) = \langle \hat{m}^{\dagger 2} \hat{m}^2 \rangle / \langle \hat{m}^\dagger \hat{m} \rangle^2$, respectively. For magnons, if $0 < g^{(2)}_m(0) < 1$, it indicates that the magnons exhibit antibunching effect, and a single magnon tends to exist in the YIG sphere. If $g^{(2)}_m(0) > 1$, the magnons exhibit a bunching effect, and multiple magnons tend to exist in the YIG sphere. The temporal correlation[31,65,78] of magnons and photons can be characterized by their cross-correlation function $g^{(2)}_{am}(0) = \langle \hat{a}^\dagger \hat{a} \hat{m}^\dagger \hat{m} \rangle / (\langle \hat{a}^\dagger \hat{a} \rangle \langle \hat{m}^\dagger \hat{m} \rangle)$.

The statistical characteristics of cross-correlation are similar to those of the single mode auto-correlation function. When $g^{(2)}_{am}(0) > 1$, it means that photons and magnons tend to exist in the system at the same time. This means that if there is a photon and a magnon in the system at the same time, these correlation functions should be $g^{(2)}_a(0) = 0$, $g^{(2)}_m(0) = 0$ and $g^{(2)}_{am}(0) > 1$. The statistical result in this case will violate the Cauchy–Schwarz inequality,[65,79,80] which is expressed as $g^{(2)}_{am}(0) \leq \sqrt{g^{(2)}_a(0) g^{(2)}_m(0)}$. Violation of the Cauchy–Schwarz inequality is a way to manifest quantum effects. We introduce a violation coefficient to represent the nonclassicality, defined as

$$R = \frac{g^{(2)}_{am}(0)}{\sqrt{g^{(2)}_a(0) g^{(2)}_m(0)}} \tag{8}$$

When $R > 1$, the Cauchy–Schwarz inequality is violated, indicating that the cross-correlation statistics are nonclassical.

Under weak driving conditions, to solve these correlation functions analytically, the state of the system can be truncated in few excitation subspace and approximately expressed as

$$\begin{aligned}|\psi\rangle &= C_{00g}|00g\rangle + C_{10g}|10g\rangle + C_{01g}|01g\rangle \\ &+ C_{00e}|00e\rangle + C_{20g}|20g\rangle + C_{02g}|02g\rangle \\ &+ C_{11g}|11g\rangle + C_{10e}|10e\rangle + C_{01e}|01e\rangle\end{aligned} \tag{9}$$

where each $|n_1 n_2 s\rangle$ and $C_{n_1 n_2 s}$ represent the possible states of the system and their probability amplitudes, respectively. n_1 and n_2 denote the number states of the photons and magnons respectively, and s denotes ground $|g\rangle$ or excited state $|e\rangle$ of the qubit. Under weak driving conditions, we have $C_{00g} \gg \{C_{10g}, C_{01g}, C_{00e}\} \gg \{C_{20g}, C_{11g}, C_{02g}, C_{10e}, C_{01e}\}$ and $C_{00g} \approx 1$. These correlation functions can be approximated by the probability amplitude as $g^{(2)}_a(0) = 2|C_{20g}|^2/|C_{10g}|^4$, $g^{(2)}_m(0) = 2|C_{02g}|^2/|C_{01g}|^4$, $g^{(2)}_{am}(0) = |C_{11g}|^2 / (|C_{10g}|^2 |C_{01g}|^2)$. Using the Schrödinger equation, the probability amplitude under steady-state conditions (the derivative of the probability amplitude with respect to time is zero) can be solved, and the magnon blockade condition can be obtained. Based on the symmetry of the system, in order to make the blockade effect of the magnon and photon modes the same, we always make the photon–qubit coupling equal to the magnon–qubit coupling, and the cavity drive equal to the magnon drive. This means that magnon–qubit coupling is indispensable. Therefore, in the absence of adiabatic elimination of cavity B, and only in cavity B (without cavity A),[12,13,21] the lack of magnon–qubit coupling makes it difficult for magnons and photons to achieve the same blockade effect. When $E_q = E_m = E$, $g_{aq} = g_{mq} = g_0$, and $g_{am} = (\Delta_a + \Delta_m)/2$, we obtain these simplified second-order correlation functions as

$$g^{(2)}_a(0) \approx \frac{|\Delta_q F_3|^2}{|(\Delta_q E - g_0 \Omega)^2 (F_1 - 4\Delta_q F_2)|^2} \tag{10}$$

$$g^{(2)}_{am}(0) \approx \frac{|\Delta_q F_4|^2}{|(\Delta_q E - g_0 \Omega)^2 (F_1 - 4\Delta_q F_2)|^2} \tag{11}$$

$$g^{(2)}_m(0) \approx g^{(2)}_a(0) \tag{12}$$

where

$$F_1 = (\Delta_a + \Delta_m)(\Delta_a - \Delta_m)^2 \tag{13}$$

$$F_2 = (\Delta_a + \Delta_m)(\Delta_a + \Delta_m + \Delta_q) - 2g_0^2 \tag{14}$$

$$F_3 = (\Delta_q E - 2g_0 \Omega)(F_1 - 4\Delta_q F_2) E - 4g_0^2 \Omega^2 F_2 \tag{15}$$

$$F_4 = F_3 - 2g_0^2 \Omega^2 (\Delta_a - \Delta_m)^2 \tag{16}$$

In this case, the condition for the magnon/photon auto-correlation to be 0 is $F_3 = 0$, which is an unconventional blockade condition. Furthermore, if $\Delta_a \neq \Delta_m$, the magnon–photon cross-correlation will not be zero, which means that there may be a strong cross-correlation.

3. Numerical Simulation and Discussions

In this section, we will discuss the blockade effects of photons and magnons using numerical simulations. Numerical simulations are conducted using the quantum master equation. The quantum master equation of the system can be written as

$$\begin{aligned}\frac{\partial \hat{\rho}}{\partial t} &= -i[\hat{H}_{\text{eff}}, \hat{\rho}] + \frac{\kappa_a}{2}(\overline{n}_a + 1)\mathscr{L}_a[\hat{\rho}] + \frac{\kappa_a}{2}\overline{n}_a \mathscr{L}_{a^\dagger}[\hat{\rho}] \\ &+ \frac{\kappa_m}{2}(\overline{n}_m + 1)\mathscr{L}_m[\hat{\rho}] + \frac{\kappa_m}{2}\overline{n}_m \mathscr{L}_{m^\dagger}[\hat{\rho}] \\ &+ \frac{\gamma}{2}(\overline{n}_q + 1)\mathscr{L}_{\sigma_-}[\hat{\rho}] + \frac{\gamma}{2}\overline{n}_q \mathscr{L}_{\sigma_+}[\hat{\rho}]\end{aligned} \tag{17}$$

where $\mathscr{L}_o[\hat{\rho}] = 2\hat{o}\hat{\rho}\hat{o}^\dagger - \hat{o}^\dagger \hat{o}\hat{\rho} - \hat{\rho}\hat{o}^\dagger \hat{o}$ $(\hat{o} = \hat{a}, \hat{a}^\dagger, \hat{m}, \hat{m}^\dagger, \hat{\sigma}_-, \hat{\sigma}_+)$ are the Lindblad superoperators,[81] and $\hat{\rho}$ is density operator of the system. $\overline{n}_a$, $\overline{n}_m$, and $\overline{n}_q$ are the thermal occupation numbers of photons, magnons, and qubit, respectively. $\overline{n}_j = [\exp(\hbar\omega_j/$

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$K_B T) - 1]^{-1}$ $(j = a,\ m,\ q)$ with the Boltzmann constant K_B and the temperature T. The expressions for calculating the correlation functions using the density matrix are $g_a^{(2)}(0) = \mathrm{Tr}(\hat{\rho}\hat{a}^{\dagger 2}\hat{a}^2)/[\mathrm{Tr}(\hat{\rho}\hat{a}^\dagger\hat{a})]^2$, $g_m^{(2)}(0) = \mathrm{Tr}(\hat{\rho}\hat{m}^{\dagger 2}\hat{m}^2)/[\mathrm{Tr}(\hat{\rho}\hat{m}^\dagger\hat{m})]^2$, and $g_{am}^{(2)}(0) = \mathrm{Tr}(\hat{\rho}\hat{a}^\dagger\hat{a}\hat{m}^\dagger\hat{m})/[\mathrm{Tr}(\hat{\rho}\hat{a}^\dagger\hat{a})\mathrm{Tr}(\hat{\rho}\hat{m}^\dagger\hat{m})]$. The numerical simulation process is to use the quantum master equation to solve the density matrix under steady-state conditions (the derivative of the density operator with respect to time is zero) and then calculate the result of the correlation function. We will combine numerical simulation with analytical calculation to analyze the quality of magnon–photon pair preparation in the system.

In Figure 1b,c, we consider the simplest case where we define the following equivalences: $\Delta_a = \Delta_m = \Delta_q = \Delta$, $\kappa_a = \kappa_m = \gamma = \kappa$, $g_{am} = g_{aq} = g_{am} = g_0$, and $E_a = E_m = E$. Under these conditions, we plotted the magnon auto-correlation function $g_m^{(2)}(0)$ and average number of magnons $N_m = \langle \hat{m}^\dagger \hat{m} \rangle$, both as functions of detuning. When the qubit driving strength Ω increases, one of the optimal points of the correlation function $g_m^{(2)}(0)$ gradually approaches another and finally coincide when $\Omega = 0.5\kappa$. The value of the correlation function $g_m^{(2)}(0)$ at this case is very low, reaching $10^{-5.8}$. These two optimal points correspond to the conventional blockade condition $\Delta = g_0$ and the unconventional blockade condition $\Delta = \frac{g_0}{4E}(4\Omega - E - \sqrt{8\Omega^2 - 15E^2 + 8\Omega E})$. The unconventional blockade condition is calculated by the probability amplitude of the two magnon state satisfying $C_{02g} = 0$ under the condition where the dissipation of the system is neglected, as this would indicate that the two magnon state $|02g\rangle$ is eliminated due to the interference of other transition paths. In general, the calculation of unconventional blockade takes into account the effect of dissipation to obtain the complete unconventional blockade condition. Since we have neglected the effect of dissipation, such calculated results are only applicable under the strong coupling condition ($g_0 \gg \kappa$). However, this incomplete unconventional blockade, when combined with conventional blockade, leads to a very low correlation function $g_m^{(2)}(0)$, indicating that the purity of the single magnon prepared under this condition is very high.[52]

In Figure 1c, we can see that with the increase of driving strength, the number of magnons at the optimal point is also increasing. This is because the driving of the qubit also has a gain effect on the number of magnons. When $\Omega = 0.5\kappa$ and $\Delta = 20\kappa$, the average number of magnons is 0.038. Conventional magnon blockade is caused by the anharmonicity of energy levels. When the driving frequency resonates with the frequency of the single magnon state $|01g\rangle$ energy level, it will be detuned with the two magnon state $|02g\rangle$ energy level, thus suppressing the excitation of the two magnon state. The quantum interference caused by the transition between energy levels further suppresses the excitation of the two magnon state. Therefore, the blockade effect is best when conventional magnon blockade and unconventional magnon blockade occur simultaneously, and the average number of magnons is relatively large. In addition, under the simplest parameter conditions, $g_m^{(2)}(0) = g_a^{(2)}(0) = g_{am}^{(2)}(0)$, which means that the cross-correlation under the optimal blockade condition is also low, and the magnons and photons are also antibunched. According to the Cauchy–Schwarz inequality violation coefficient $R = 1$ indicates that there is no quantum effect between magnons and photons at this case. Therefore, both magnon blockade and photon blockade can be satisfied, but photon and magnon do not tend to appear in pairs, so we can only obtain single magnons or single photons, respectively.

In order to study the conditions for having a strong cross-correlation, we consider the case where the detuning Δ_a, Δ_m, and Δ_q take different parameters. We found through analysis that when $g_0 = (\Delta_a + \Delta_m)/2$, the $g_a^{(2)}(0)$ is approximately equal to the $g_m^{(2)}(0)$, hence they share the same region where the value is less than 0. In **Figure 2**a,b, we have fixed the magnon detuning $\Delta_m = -100\kappa$, so the coupling strength g_0 varies with the cavity detuning Δ_a. We plotted the $g_m^{(2)}(0)$ and the $g_{am}^{(2)}(0)$ separately,

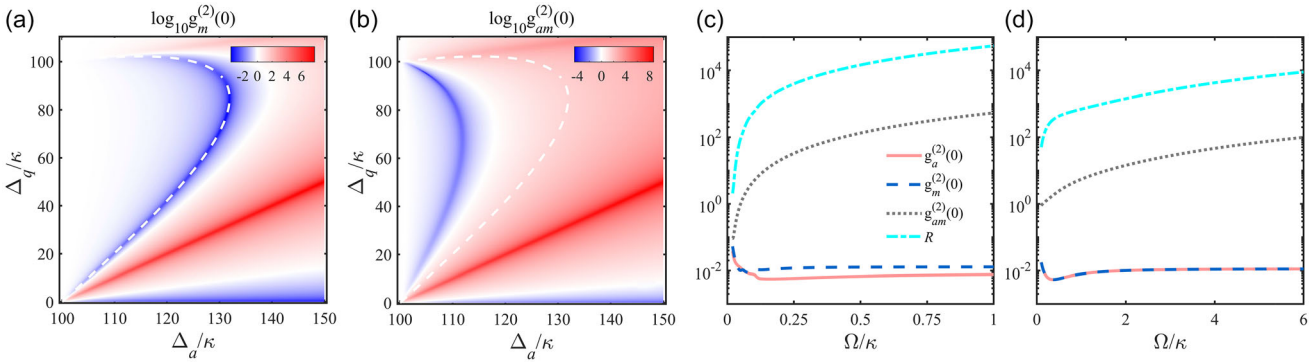

Figure 2. The logarithmic magnon auto-correlation $\log_{10} g_m^{(2)}(0)$ and the magnon–photon cross-correlation function $\log_{10} g_{am}^{(2)}(0)$ are shown in a,b), respectively, as a function of the detuning Δ_a and Δ_q. The fixed magnon detuning is $\Delta_m = -100\kappa$, and the coupling strength satisfies $g_0 = (\Delta_a + \Delta_m)/2$. The qubit driving strength is $\Omega = 0.3\kappa$, and the other parameters are the same as Figure 1b. The white dashed line is the unconventional blockade condition Equation (18). The $g_a^{(2)}(0)$, $g_m^{(2)}(0)$, $g_{am}^{(2)}(0)$ and R as a function of the qubit driving strength Ω. c) In the weak driving region, the coupling strength g_0 is fixed at 5κ, the magnon detuning is $\Delta_m = 2g_0 - \Delta_a$, the qubit detuning is $\Delta_q = -2\Delta_m/3$, and the cavity detuning Δ_a satisfies Equation (18). The driving strength $E = 0.1\kappa$. d) In the strong qubit driving region, the coupling strength g_{am}, the qubit detuning Δ_q, and the driving strength E are all fixed at 0. Coupling strengths g_{aq} and g_{mq} are both 0.1κ, the cavity detuning is $\Delta_a = 40\,\Omega$, and the magnon detuning is equal to the negative of the cavity detuning $\Delta_m = -\Delta_a$.

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and there are distinct differences between them. We can observe that the dark blue section in Figure 2a is the region where $g_m^{(2)}(0)$ is less than 0, indicating that unconventional blockade has occurred under these conditions. The analysis result of unconventional blockade is

$$0 = (\Delta_q - g_0\eta)^2(\Delta_q + g_0) - (\Delta_q - 2g_0\eta)(\Delta_a - g_0)^2 \tag{18}$$

where $\eta = \Omega/E$. We have marked the unconventional blockade condition with a white dashed line in Figure 2a,b. The white dashed line coincides with the dark blue region in Figure 2a, but does not coincide with the dark blue region in Figure 2b ($g_{am}^{(2)}(0) < 0$). In Figure 2b, the position of the white dashed line corresponds to $g_{am}^{(2)}(0)$ greater than 1, which implies that magnons and photons tend to appear in pairs.

In Figure 2c, we fixed the coupling strength $g_0 = 5\kappa$. Under the unconventional blockade condition Equation (18), the magnon–photon cross-correlation function $g_{am}^{(2)}(0)$ is significantly greater than 1. As the qubit driving strength Ω continuously increases, $g_{am}^{(2)}(0)$ is also enhanced, while the auto-correlation functions $g_a^{(2)}(0)$ and $g_m^{(2)}(0)$ are close to 0.01. This indicates that, to a certain extent, enhancing the qubit driving strength does not suppress the blockade effect but can improve the cross-correlation. According to the definition of the R parameter, when the qubit driving strength is $\Omega = 0.2\kappa$, R exceeds 10^3. This value clearly exceeds the threshold for violating the Cauchy–Schwarz inequality, indicating that the strong magnon–photon cross-correlation observed under unconventional blockade conditions is a distinctively quantum phenomenon. Compared with[65] adding qubit drive can make this scheme show better cross-correlation effect and quantum properties.

In Figure 2d, we consider the case where only the qubit is driven, with the qubit driving strength exceeding the dissipation of the qubit γ, but the coupling strengths g_{aq} and g_{mq} are lower than the dissipation γ. Under strong qubit driving conditions, when $\Delta_q = 0$, the probability of the state $|00e\rangle$ will significantly increase, leading to both the $|00g\rangle$ and $|00e\rangle$ probabilities being approximately 0.5. The weak coupling strengths result in new probabilities presented as $\{C_{00g}, C_{00e}\} \gg \{C_{10g}, C_{10e}, C_{01g}, C_{01e}\} \gg \{C_{20g}, C_{20e}, C_{11g}, C_{11e}, C_{02g}, C_{02e}\}$. At this case, the coupling is equivalent to driving, and the driving is equivalent to coupling. When $\Delta_m = -\Delta_a$ and $\kappa_a = \kappa_m$, the real parts of C_{10g} and C_{01g} are opposite, while their imaginary parts are identical. Similarly, analogous conclusions hold for C_{20g} and C_{02g}, resulting in the auto-correlation functions $g_m^{(2)}(0)$ and $g_a^{(2)}(0)$ being equal, hence they completely overlap. The probability of the state $|11g\rangle$ is resonantly enhanced, which is due to the detuning that suppresses other multi-magnon/photon states. This results in a strong magnon–photon cross-correlation and a weak magnon/photon auto-correlation, indicating a strong tendency for the system to simultaneously contain one photon and one magnon.

Under different blockade conditions, changing these dissipations has different effects on these three correlation functions. From **Figure 3**a, we can clearly see that the blockade effect is best only when $\kappa_a = \kappa_m$. This indicates that when $\kappa_a \neq \kappa_m$, the unconventional photon (magnon) blockade condition will not be satisfied, that is, C_{20g} (C_{02g}) is not equal to 0. Because we assumed $\Delta_a = \Delta_m$ when calculating the optimal single photon (magnon) blockade conditions, the dissipations κ_a and κ_m also need to be equal. Figure 3b shows that as the qubit dissipation γ increases, these three correlation functions gradually increase. This is because when the dissipation rate of the qubit increases, the population of the excited state $|e\rangle$ will decrease. And due to the coupling transition between energy levels $|00e\rangle$ and $|10g\rangle$ ($|01g\rangle$), the probability amplitude C_{10g} (C_{01g}) is reduced, so the purity of the prepared photons (magnons) will also decrease. The dissipation of qubits can reach $\gamma/2\pi = 50$ kHz,[82] which is much lower than the dissipation of magnons at $\kappa_m/2\pi = 1$ MHz, therefore enabling the realization of high-purity photon (magnon) sources.

As shown in Figure 3c,d, under the optimal conditions for strong cross-correlation, the auto-correlation is lower when the dissipation is lower. This is because under this condition, the large detuning to avoid unconventional blockade is no longer satisfied due to the change in dissipation, but the increase in dissipation still leads to a decrease in purity. Unlike the previous case, as shown in Figure 3e, under strong qubit driving conditions, when the cavity dissipation κ_a increases, the magnon–photon cross-correlation decreases significantly, and the photon auto-correlation also decreases. Since the qubit detuning Δ_q is 0 under this condition, the corresponding excited state $|10e\rangle$ is more significantly affected by the dissipation κ_a. And the $|20g\rangle$ and $|11g\rangle$ states with coupling transitions to the energy level $|10e\rangle$, both C_{20g} and C_{11g}, will also decrease with the increase in dissipation κ_a. Therefore, photon auto-correlation and magnon–photon cross-correlation decrease at the same time. The changes in auto-correlation in Figure 3f are similar to Figure 3b and for the same reason.

Due to the lower frequency of magnons, they are more susceptible to thermal noise interference. In Figure 3g–i, we assume the frequency of the magnons, cavity A, and qubit equal to $\omega_m/2\pi = 8.5$ GHz. As the temperature increases, the thermal occupation numbers of magnons and photons also increase, leading to state distribution of the system to be more in the thermal statistical distribution. This disrupts the original state distribution under the blockade condition, ultimately causing the blockade effect to completely disappear. We can observe that the auto-correlation functions $g_a^{(2)}(0)$ and $g_m^{(2)}(0)$ eventually approach 2, while the cross-correlation function $g_{am}^{(2)}(0)$ eventually approaches 1. This is because the second-order auto-correlation statistical result of a pure thermal state is 2, and the cross-correlation statistical result of two uncorrelated thermal states is 1. This fully indicates that when the thermal occupation numbers of magnons and photons are sufficiently large, thermal noise will destroy both the blockade effect and the strong cross-correlation effect. However, as shown in Figure 3i, when the temperature is below 18 mK, the statistical results of magnons and photons do not show significant changes. Since such low temperatures have been achieved in experiments,[5,12] discussions on magnon/photon blockade and magnon–photon strong cross-correlation can be realized under experimental conditions.

Later, we analyze the experimental feasibility of the scheme in this article. We think that the main challenge is whether each

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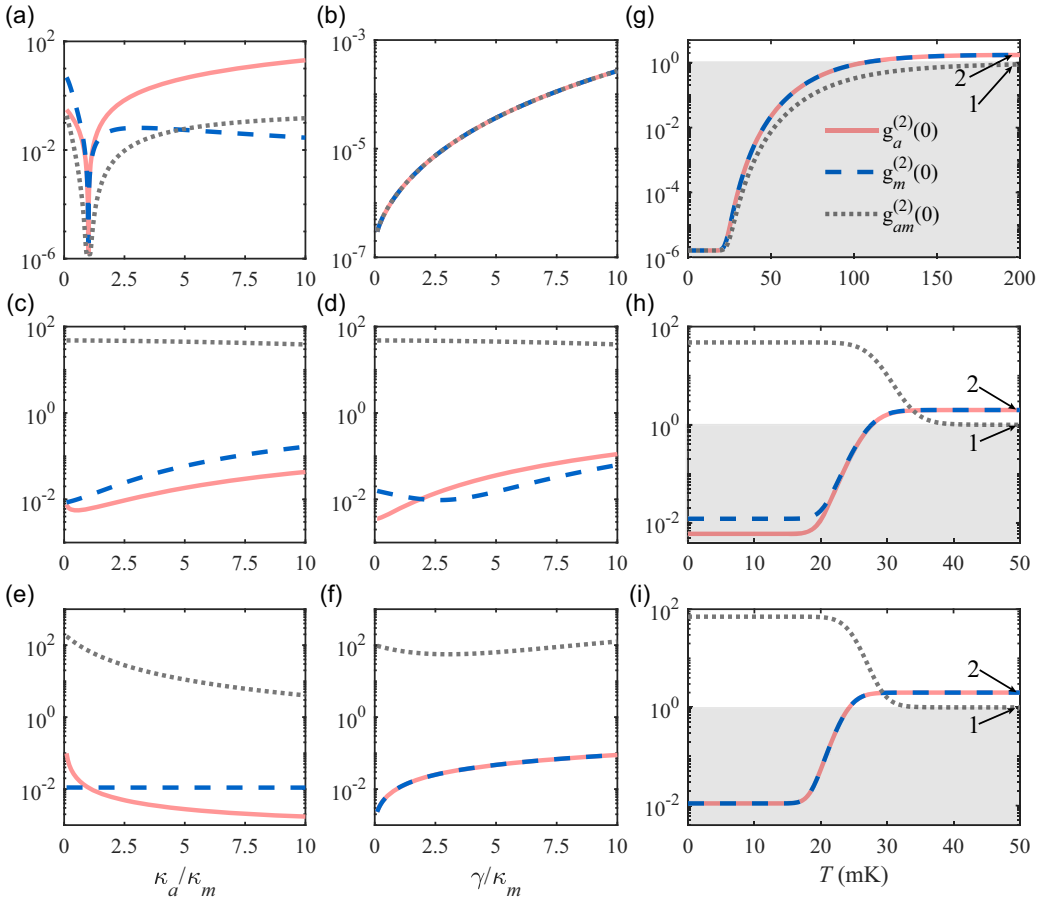

Figure 3. The $g_a^{(2)}(0)$, $g_m^{(2)}(0)$, and $g_{am}^{(2)}(0)$ as a function of cavity A dissipation κ_a and qubit dissipation γ, respectively. Figure a–f) select the parameters when $\Omega = 0.5\kappa$ and $\Delta = 20\kappa$ in Figure 1b, $\Omega = 0.3\kappa$ in Figure 2c, and $\Omega = 5\kappa$ in Figure 2d, respectively. The $g_a^{(2)}(0)$, $g_m^{(2)}(0)$, and $g_{am}^{(2)}(0)$ as a function of the temperature T. Figure g–i) respectively select the parameters when $\Omega = 0.5\kappa$ and $\Delta = 20\kappa$ in Figure 1b, $\Omega = 0.3\kappa$ in Figure 2c, and $\Omega = 5\kappa$ in Figure 2d, respectively.

part can achieve a sufficiently strong coupling. In ref. [83] a rectangular 3D microwave cavity has a detuning of $\Delta/2\pi \approx 0.5$ GHz, the cavity–magnon coupling strength is $g_m/2\pi \approx 42$ MHz, and the effective coupling between the magnon and the qubit is $g_{mq}/2\pi \approx 20.1$ MHz. According to the effective coupling relationship $g_m = g_m g_q/\Delta$, the qubit-magnon coupling strength is calculated to be $g_q/2\pi \approx 239$ MHz. In the theoretical paper,[14] it is pointed out that magnons and qubits can be directly coupled through a changing magnetic flux, with a coupling strength of up to $g_{mq}/2\pi \approx 35$ MHz. Although, up to now, we have not found any experimental articles where YIG spheres, microwave cavities, and qubits are all directly coupled, we believe that a system coupled in this way is feasible. We think that the currently feasible solution is to implement it within a superconducting circuit. In ref. [82] the coupling between two microwave cavities reaches $g_{ab}/2\pi \approx 249$ MHz, and the coupling between the microwave cavity and the qubit reaches $g_{bq}/2\pi \approx 311$ MHz. Moreover, the nonradiative decay rates for the qubit and the microwave cavity are $\gamma/2\pi = 50$ kHz and $\kappa_a/2\pi = 300$ kHz, respectively. In ref. [84] the coupling between the magnon and photon reaches $g_{bm}/2\pi \approx 130$ MHz, with both the magnon and photon damping rates approaching 1 MHz.

4. Conclusion

In conclusion, we have theoretically investigated the conditions for achieving strong second-order cross-correlation between magnons and microwave photons within a cavity–magnon–qubit system. The results demonstrate that such strong cross-correlation can be achieved under unconventional photon blockade conditions. When both conventional magnon blockade and unconventional magnon blockade are achieved in the system, the magnons have a very low auto-correlation and a large average number of magnons, which indicates that the prepared magnon state has a high purity and efficiency. Since the cross-correlation function between magnons and photons is also very low when these two conditions are met at the same time, the Cauchy–Schwarz inequality violation coefficient is equal to 1 in this case, so there is no quantum property between magnons and photons, and they can only be regarded as a single magnon or photon source. However, the strong magnon–photon cross-correlation achieved only under unconventional blockade conditions has a Cauchy–Schwarz inequality violation coefficient significantly exceeding 1, indicating that there is strong quantum effect between magnons and photons, and single magnons and single

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photons can be prepared in the system at the same time. This scheme can use the blockade effect to prepare strongly correlated magnon–photon pairs. Moreover, the switching of magnons and photons from correlation to non-correlation can be achieved by adjusting parameters. This scheme provides a promising approach for preparing hybrid quantum states, which is expected to have potential applications in quantum information processing and quantum sensing technology.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Yi Ren: investigation (lead); validation (lead); and writing—original draft (lead). **Kai Wei Huang**: methodology (supporting) and writing—review and editing (supporting). **Xin Wang**: methodology (supporting) and writing—review and editing (supporting). **Hao Xiong**: conceptualization (lead); methodology (lead); and writing—review and editing (lead).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

magnon–photon pair, photon blockade, quantum correlation

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